

SIYU YANG

Houston, TX — Siyu.Yang@uth.tmc.edu — (313)266-8072 — siyu-yang.github.io/

RESEARCH INTERESTS

My research interests lie broadly in **AI for biomedical/healthcare**, with a focus on:

- **Generative AI for biomedicine & therapeutics**: Sequence-structure co-design, conditional generation, and optimization algorithms for therapeutic biomolecules.
- **Multimodal Learning**: Integrating heterogeneous biological and clinical data modalities for predictive and generative healthcare tasks.
- **Geometric deep learning and topological learning**: Developing topological and geometric representation learning frameworks for complex biological systems, interactions, and health networks.

EDUCATION

University of Texas Health Science Center at Houston

M.S. in Biomedical Informatics – Research Track

Houston, TX

Expected Dec 2026

University of Notre Dame

M.S. in Computer Science and Engineering

Notre Dame, IN

May 2025

University of Electronic Science and Technology of China (UESTC)

B.E. in Computer Science and Technology & B.S. in Finance

Chengdu, China

Jun 2020

RESEARCH EXPERIENCE

UTHealth Houston

Graduate Research Assistant, AI for Antibody Design Team

Advisor: Prof. Yejin Kim

Houston, TX

Sep 2025 – Present

- Developed end-to-end **generative AI framework** for *de novo* antibody design against transmembrane therapeutic targets, integrating generative design, sequence optimization, structure prediction and evaluation.
 - Designed **constrained hallucination** and multi-objective optimization strategies to generate epitope-targeted antibodies, leveraging structure predictors and protein language models (**PLMs**) to ensure structural stability and high sequence naturalness.
 - Performed structure-aware sequence optimization of antibody complementarity-determining regions (CDRs) via **inverse folding** approaches (e.g., ProteinMPNN).
 - Engineered robust **multimodal evaluation and ranking frameworks** by integrating deep-learning confidence metrics (e.g., AlphaFold 3 outputs) with domain-specific physical constraints to guide high-throughput screening.
- **Current Research**: Exploring representation learning and constraint-guided generative modeling for antibody design, incorporating substructural features and transmembrane specific priors to improve binding affinity and target specificity.

University of Notre Dame

Graduate Research Assistant, Complex Networks Lab

Advisor: Prof. Tijana Milenković

Notre Dame, IN

Sep 2022 – Aug 2024

- Leveraged **network-based** methods to analyze biological networks, capturing higher-order topological substructures and enabling analysis of both protein structural properties and protein interaction patterns.

- Constructed **dynamic Protein Structure Networks** (PSNs) to adaptively represent protein structures over time and identify critical structural changes.
- Developed a multi-layered **Network of Networks** (NoN) hierarchical framework by integrating Protein-Protein Interaction (PPI) networks and dynamic PSNs. Applied the framework to protein function prediction, achieving improved performance compared to static network-based approaches.
- Explored graph-based representations of proteins, building a foundation for machine learning approaches in protein modeling.

University of Michigan-Dearborn

Research Assistant, Intelligent Systems Lab

Advisor: Prof. Yi Lu Murphey

Dearborn, MI

Jan 2021 – Aug 2022

- Developed deep learning models for large-scale time-series forecasting using real-world vehicle data.
- Designed and implemented algorithms for object recognition (stop signs, traffic lights) and developed a method to segment and categorize roadways using data integration techniques.

PUBLICATIONS & PRESENTATIONS

- “Dynamic network analysis of multi-scale-omics data for protein function prediction”, **S. Yang** (presenter), T. Milenković, **selected as oral presentation**, ISMB’24, July 12–16, 2024, Montreal, Canada.
- “Dynamic network analysis of protein structural change”, A. Wells (presenter), **S. Yang**, K. Newaz, T. Milenković, **selected as oral presentation**, ISMB’24, July 12–16, 2024, Montreal, Canada.
- Z. Zhao, **S. Yang**, C. Sauer, A. Teraji, Y. Yamauchi, T. Hirata, A.J. Bisci, Y.L. Murphey, “Structured Deep Learning Models for Accurate Prediction of Real-world Driving Speed for Short and Long-term Horizons”, *IJCNN*, 2022.

TEACHING & SERVICE

Teaching Assistant:

CSE-30151 Theory of Computing, University of Notre Dame, 23FA, 24SP, 24FA, 25SP

CIS-450 Operating Systems, University of Michigan-Dearborn, 20SP

Student Volunteer: ISMB 2024

AWARDS & HONORS

- Student Travel Fellowship, ISMB, 2024
- Overseas Exchange Scholarship, UESTC, 2020
- Best Student Assistant, UESTC, 2019
- Third Class Undergraduate Scholarship, UESTC, 2018

TECHNICAL SKILLS

Core Development & Programming: Python, Julia, R, C/C++, SQL, PyTorch, PyG, NetworkX, Linux/Bash, SLURM, Docker, Git, Hugging Face, PySpark.

GenAI for Biomedical: BindCraft, ColabDesign, RFdiffusion, ProteinMPNN, AlphaFold, ChimeraX.